



Lecture Subject

An Introduction to Research Methodology

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March 2022

What is Research?

From the French word "**recherche**" which means to travel through or survey



Research is an active, diligent and systematic process of inquiry in order to **discover**, **interpret** or **revise facts, events, behaviors, or theories**, or to make practical applications with the help of such facts, laws or theories. The term 'research' is also used to describe the collection of information about a particular subject

What is Research?

(Some other Insightful Definitions)

- A process of enquiry and investigation; It is systematic, methodical and ethical; Research can help solve practical problems and increase knowledge.
- Systematic, intensive, patient study and investigation in some field of knowledge, usually employing the techniques of hypothesis and experiment, whose purpose is to reveal new facts, theories, or principles
- A systematic investigation, including research development, testing and evaluation, designed to develop or contribute to generalizable knowledge
- Systematic study directed toward more complete scientific knowledge or understanding of the subject studied

What is Research?

(Some fundamental concepts)

- Research follows a scientific method.
- This means that it makes an integrated use of reasoning.
- This makes it very useful for explaining and/or predicting phenomena.
- The basic assumption of the scientific method is that every effect has a cause.

The Purpose of Research

- Explain new phenomenon
- Generate new knowledge
- Provide solutions to problems
- Explore and analyze problems
- Review or synthesize existing knowledge
- Investigate existing situations or problems
- Construct or create new procedures or systems

High Quality Research

- It is doable!
- It generates new questions.
- It is based on the work of others.
- It can be replicated (duplicated).
- It is generalizable to other settings.
- It is based on some logical rationale and tied to theory.

Motivations of Research

- To solve the problems
- To improve our life
- To make new discoveries
- To investigate laws of nature
- To provide solutions to complex problems
- To develop new products
- To get PhDs, Masters, Scholars, Reputation and Funds
- Human desires/intellectual joy

Types of Research

- 1) Fundamental
- 2) Exploratory
- 3) Applied
- 4) Descriptive
- 5) Analytical
- 6) Predictive

1) Fundamental Research

- Mainly concerned with generalizations and with the **formulation of a theory**
- Research concerning some **natural phenomenon** or relating to pure mathematics are examples of fundamental research
- Basic research is directed towards finding information that has a broad base of applications
- Adds to the already existing organized body of scientific knowledge

2) Exploratory Research

- Exploratory research is undertaken when few or no previous studies exist.
- The aim is to look for patterns, hypotheses or ideas that can be tested and will form the basis for further research.
- Typical research techniques would include case studies, observation and reviews of previous related studies and data.

3) Applied Research

- Aims at finding a solution for an immediate problem facing a society or an organization
- To discover a solution for some pressing practical problem

4) Descriptive Research

- Descriptive research can be used to identify and classify the elements or characteristics of the subject

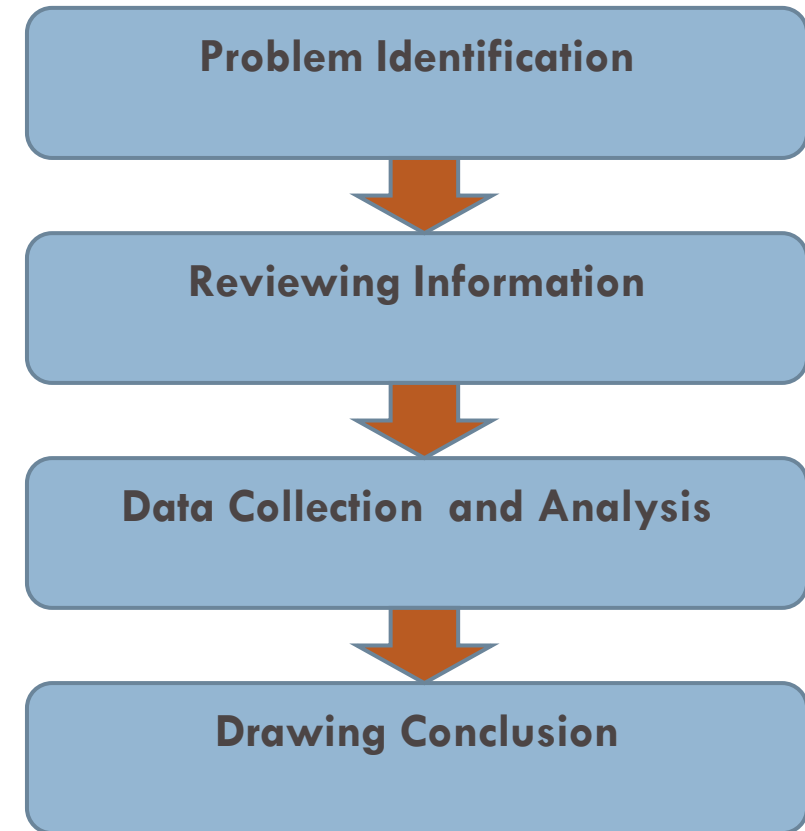
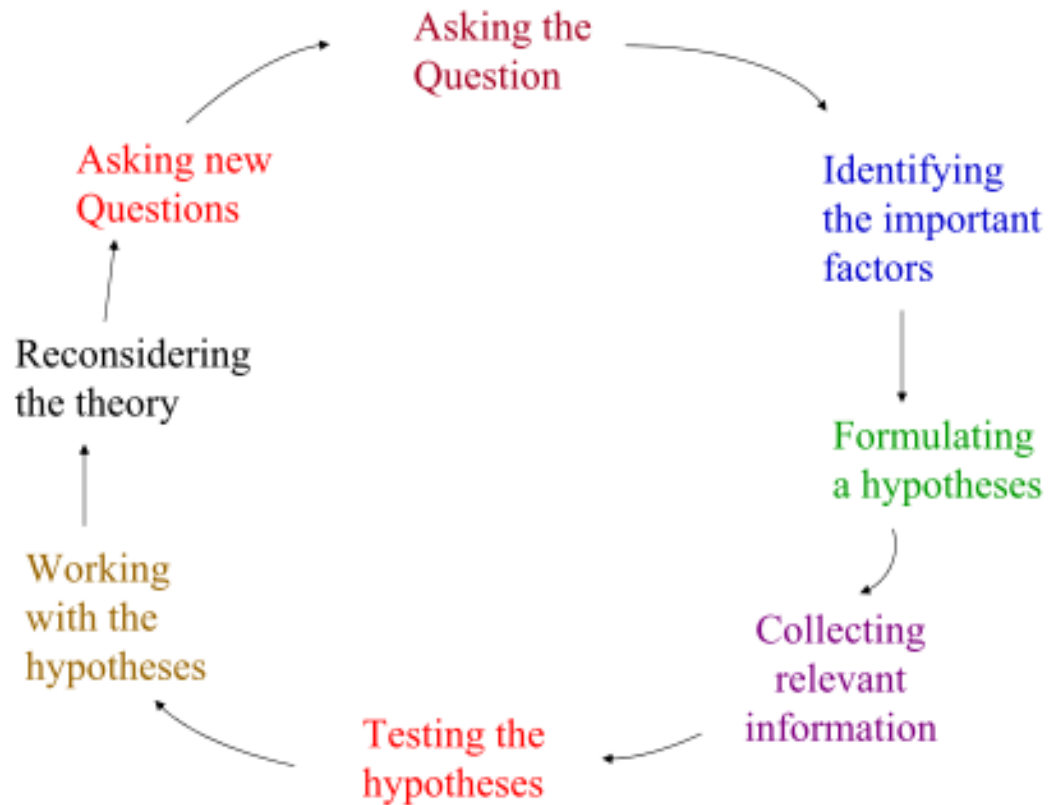
5) Analytical Research

- Analytical research often extends the Descriptive approach to suggest or explain why or how something is happening

6) Predictive Research

- The aim of Predictive research is to speculate intelligently on future possibilities, based on close analysis of available evidence of cause and effect

Steps of Conducting a Research



Some Research Example

- Systematic Literature Review (SLR)
- Analytical / Applied Research
- Dataset Publish Research
- Survey

Systematic Literature Review (SLR)

- 1) Abstract
- 2) Introduction
- 3) Review of Related Studies
- 4) Research Method
- 5) Analytical Questions
- 6) Taxonomy
- 7) Summary Table of Investigated Works
- 8) Discussion and Comparison
- 9) References

Systematic Literature Review (SLR)

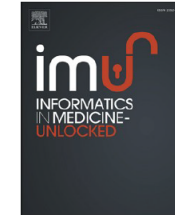
Informatics in Medicine Unlocked 23 (2021) 100550



Contents lists available at [ScienceDirect](https://www.sciencedirect.com)

Informatics in Medicine Unlocked

journal homepage: <http://www.elsevier.com/locate/imu>



Deep learning applications for IoT in health care: A systematic review

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ARTICLE INFO

Keywords:

Deep learning
Internet of things
Healthcare
Medical imaging
Wearable device
Systematic literature review

ABSTRACT

In machine learning, deep learning is the most popular topic having a wide range of applications such as computer vision, natural language processing, speech recognition, visual object detection, disease prediction, drug discovery, bioinformatics, biomedicine, etc. Of these applications, health care and medical science-related applications are dramatically on the rise. The tremendous big data growth, the Internet of Things (IoT), connected devices, and high-performance computers utilizing GPUs and TPUs are the main reasons why deep learning is so popular. Based on their specific tasks, medical IoT, digital images, electronic health record (EHR) data, genomic data, and central medical databases are the primary data sources for deep learning systems. Several potential issues such as privacy, QoS optimization, and deployment indicate the pivotal part of deep learning. In this paper, deep learning for IoT applications in health care systems is reviewed based on the Systematic Literature Review (SLR). This paper investigates the related researches, selected from among 44 published research papers, conducted within a period of ten years – 2010 to 2020. Firstly, theoretical concepts and ideas of deep learning and technical taxonomy are proposed. Afterwards, major deep learning applications for IoT in health care and medical sciences are presented through analyzing the related works. Later, the main idea, advantages, disadvantages, and limitations of each study are discussed, preceding suggestions for further research.



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journal homepage: www.journals.elsevier.com/medicine-in-novel-technology-and-devices/



Review Article

Depression diagnosis by deep learning using EEG signals: A systematic review



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ARTICLE INFO

Keywords:

Deep learning
Depression
Electroencephalogram
EEG
Depressive disorders
Systematic literature review

ABSTRACT

Depression is considered by WHO as the main contributor to global disability and it poses dangerous threats to approximately all aspects of human life, in particular public and private health. This mental disorder is usually characterized by considerable changes in feelings, routines, or thoughts. With respect to the fact that early diagnosis of this illness would be of the critical importance in effective treatment, some developments have occurred in the purpose of depression detection. EEG signals reflect the working status of the human brain which are considered the most proper tools for a depression diagnosis. Deep learning algorithms have the capacity of pattern discovery and extracting features from the raw data which is fed into them. Owing to this significant characteristic of deep learning, recently, these methods have intensely utilized in the diverse research fields, specifically medicine and healthcare. Thereby, in this article, we aimed to review all papers concentrated on using deep learning to detect or predict depressive subjects with the help of EEG signals as input data. Regarding the adopted search method, we have finally evaluated 22 articles between 2016 and 2021. This article which is organized according to the systematic literature review (SLR) method, provides complete summaries of all exploited studies and compares the noticeable aspects of them. Moreover, some statistical analyses have been performed to gain a depth perception of the general ideas of the latest pieces of research in this area. A pattern of a five-step procedure has also been established by which almost all reviewed articles have fulfilled the goal of depression detection. Finally, open issues and challenges in this way of depression diagnosis or prediction and suggested works as the future directions have been discussed.

Analytical / Applied Research

- 1) Abstract
- 2) Introduction
- 3) Related Works
- 4) Fundamental Concepts
- 5) Proposed Method
- 6) Simulation / Implementation Results
- 7) Discussion and Future Works (Optional)
- 8) Conclusion
- 9) References

RESEARCH

Open Access

Deep learning accelerators: a case study with MAESTRO



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Abstract

In recent years, deep learning has become one of the most important topics in computer sciences. Deep learning is a growing trend in the edge of technology and its applications are now seen in many aspects of our life such as object detection, speech recognition, natural language processing, etc. Currently, almost all major sciences and technologies are benefiting from the advantages of deep learning such as high accuracy, speed and flexibility. Therefore, any efforts in improving performance of related techniques is valuable. Deep learning accelerators are considered as hardware architecture, which are designed and optimized for increasing speed, efficiency and accuracy of computers that are running deep learning algorithms. In this paper, after reviewing some backgrounds on deep learning, a well-known accelerator architecture named MAERI (Multiply-Accumulate Engine with Reconfigurable interconnects) is investigated. Performance of a deep learning task is measured and compared in two different data flow strategies: NLR (No Local Reuse) and NVDLA (NVIDIA Deep Learning Accelerator), using an open source tool called MAESTRO (Modeling Accelerator Efficiency via Spatio-Temporal Resource Occupancy). Measured performance indicators of novel optimized architecture, NVDLA shows higher L1 and L2 computation reuse, and lower total runtime (cycles) in comparison to the other one.

Keywords: Deep learning, Convolutional neural networks, Deep neural networks, Hardware accelerator, Deep learning accelerator

Dataset Publish Papers

- 1) Abstract
- 2) Introduction
- 3) Related Works
- 4) Proposed Dataset
- 5) Value of the Data
- 6) Discussion and Future Works
- 7) References



A histopathological image dataset for grading breast invasive ductal carcinomas

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ARTICLE INFO

Keywords:

Breast cancer
Invasive ductal carcinoma
Histopathology
Digital pathology
Grading
Image dataset

ABSTRACT

Breast cancer is a common cancer in women, and one of the major causes of death among women around the world. Invasive ductal carcinoma (IDC) is the most widespread type of breast cancer with about 80% of all diagnosed cases. Early accurate diagnosis plays an important role in choosing the right treatment plan and improving survival rate among the patients. In recent years, efforts have been made to predict and detect all types of cancers by employing artificial intelligence. An appropriate dataset is the first essential step to achieve such a goal. This paper introduces a histopathological microscopy image dataset of 922 images related to 124 patients with IDC. The dataset has been published and is accessible through the web at: <http://databio.com>. The distinctive feature of this dataset as compared to similar ones is that it contains an equal number of specimens from each of three grades of IDC, which leads to approximately 50 specimens for each grade.

1. Introduction

Cancer is a serious public health issue worldwide and the second leading cause of death in the United States [1]. According to the International Agency for Research on Cancer (IARC), about 18.1 million new cases and 9.6 million deaths caused by cancer were reported in 2018 [2].

As shown in Fig. 1, breast cancer is a common cancer and one of the major causes of death worldwide with 627,000 deaths among 2.1 million diagnosed cases in 2018 [2–6].

About 80% of all diagnosed breast cancers are invasive ductal carcinoma (IDC) which makes it the most widespread phenotypic subtype of all breast cancers [7]. It grows in milk ducts and invades fibrous tissue of the breast outside the duct. Pathologists recognize the cancer type and grade through visual investigation of tissue stained by hematoxylin and eosin (H&E) [17] as shown in Fig. 2. Accurate identification of IDC grade is essential in choosing the proper treatment plan and the outcome of the patient accordingly.

Histologic grade is a prognostic factor and an indicator of response to chemotherapy. It has been shown to be closely related to the frequency of recurrence and death due to IDC, and to disease-free interval and to longer life after mastectomy.

Several studies show that patients with a high grade IDC, and

mastectomy treated, had a remarkably higher frequency of auxiliary lymph node (ALN) and mortality rate than the patients with lower grade. Histologic grade has been shown to be significantly related not only to recurrence and death due to invasive ductal carcinoma, but also to the disease-free interval and overall length of survival after mastectomy regardless of clinical stage. High-grade carcinomas result in early treatment failures, whereas later recurrences are more often observed among low-grade tumors [8,9].

In the recent years, efforts have been made to predict and detect all types of cancers, employing artificial intelligence and its sub categories like machine learning and deep learning [10–15]. The appropriate dataset is the first essential step to achieve this goal. There are some available datasets for IDC [11–13] but our findings indicate that there is no published dataset for histologic grading of IDC. Therefore, this research aimed at providing a well-organized histopathological microscopy image dataset for grading IDC. The images have been taken from breast tissues stained with H&E and are labeled based on their grade and magnification level.

This paper has been organized as follows: Related works are presented in section 2. Section 3 covers the main contribution of this research, i.e. the prepared image dataset: databio. Methods of collecting images and doing the statistics are presented in this section.

A Dataset for Breast Cancer Histopathological Image Classification

Fabio A. Spanhol*, Luiz S. Oliveira, Caroline Petitjean, and Laurent Heutte

Abstract—Today, medical image analysis papers require solid experiments to prove the usefulness of proposed methods. However, experiments are often performed on data selected by the researchers, which may come from different institutions, scanners, and populations. Different evaluation measures may be used, making it difficult to compare the methods. In this paper, we introduce a dataset of 7909 breast cancer histopathology images acquired on 82 patients, which is now publicly available from <http://web.inf.ufpr.br/vri/breast-cancer-database>. The dataset includes both benign and malignant images. The task associated with this dataset is the automated classification of these images in two classes, which would be a valuable computer-aided diagnosis tool for the clinician. In order to assess the difficulty of this task, we show some preliminary results obtained with state-of-the-art image classification systems. The accuracy ranges from 80% to 85%, showing room for improvement is left. By providing this dataset and a standardized evaluation protocol to the scientific community, we hope to gather researchers in both the medical and the machine learning field to advance toward this clinical application.

Index Terms—Breast cancer, histopathology, image classification, medical imaging.

I. INTRODUCTION

CANCER is a significant public health problem in the world today. According to the International Agency for Research on Cancer of the World Health Organization, 8.2 million deaths were caused by cancer in 2012 and 27 million of new cases of this disease are expected before 2030 [1]. In particular, breast cancer (BC) is one of most common types of cancer among women. Mortality of BC is very high when compared to other types of cancer.

Detection and diagnosis of BC can be achieved by imaging procedures such as diagnostic mammograms (X-rays), magnetic resonance imaging, ultrasound (sonography), and thermography [2]. Imaging for cancer screening has been investigated for more than four decades [3]. However, biopsy is the only way to diagnose with confidence if cancer is really present. Among biopsy techniques, the most common are fine needle aspiration, core needle biopsy, vacuum-assisted, and surgical (open) biopsy

(SOB) [4]. The procedure consists in collecting samples of cells or tissue, which are fixed across a glass microscope slide for subsequent staining and microscopic examination. Diagnosis from a histopathology image is thus the gold standard in diagnosing almost all types of cancer, including BC [5], [6]. The final BC diagnosis, including grading and staging, is done by pathologists applying visual inspection of histological samples under microscope.

Histopathological analysis is a highly time-consuming specialized task, dependent on the experience of the pathologists and influenced by factors such as fatigue and decrease of attention. As pointed by Gurcan *et al.* [7], there is a pressing need for computer-assisted diagnosis (CAD) to relieve the workload on pathologists by filtering obviously benign areas, so that the experts can focus on the more difficult-to-diagnose cases [8].

A considerable amount of efforts has thus been devoted to the field of BC histopathology image analysis, and in particular to the automated classification of benign or malignant images, for computer-aided diagnosis. Kowal *et al.* [9] compare and test different algorithms for nuclei segmentation on a dataset of 500 images, for which accuracies ranging from 96% to 100% are reported. Filipczuk *et al.* [10] present a BC diagnosis system based on the analysis of cytological images of fine needle biopsies, to discriminate the images as either benign or malignant. Using four different classifiers trained with a 25-D feature vector, they report a performance of 98% on 737 images. Similarly to [9] and [10], George *et al.* [11] propose a diagnosis system for BC based on the nuclei segmentation of cytological images. Using different machine learning models, such as neural networks and support vector machines (SVMs), they report accuracy rates ranging from 76% to 94% on a dataset of 92 images. Zhang *et al.* [12] propose a cascade approach with rejection option. In the first level of the cascade, authors expect to solve the easy cases, while the hard ones are sent to a second level where a more complex pattern classification system is used. They assess the proposed method on a database proposed by the Israel Institute of Technology, which is composed of 361 images (40× magnification). On this dataset, they report results of 97% of reliability. In another work [13], the same authors assessed an ensemble of one-class classifiers on the same database achieving a recognition rate of 92%.

We can gather from the literature that most of the works on BC histopathology image analysis are carried out on small datasets, which are usually not available to the scientific community. In a recent review, Veta *et al.* [14] point out that the main obstacle in the development of new histopathology image analysis methods is the lack of large, publicly available, annotated datasets. Annotated database is also crucial to develop and validate machine learning systems.

Manuscript received May 14, 2015; revised August 18, 2015 and October 21, 2015; accepted October 22, 2015. Date of publication October 30, 2015; date of current version June 16, 2016. This work was supported by the National Council for Scientific and Technological Development (CNPq), Brazil, under grant #301653/2011-9 and the Coordination for the Improvement of Higher Level Personnel (CAPES) through SticAmsud project #263/2014. Asterisk indicates corresponding author.

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Color versions of one or more of the figures in this paper are available online at <http://ieeexplore.ieee.org>.

Digital Object Identifier 10.1109/TBME.2015.2496264

Date of publication xxxx 00, 0000, date of current version xxxx 00, 0000.

Digital Object Identifier 10.1109/ACCESS.2020.3039858

Hardware and Software Optimizations for Accelerating Deep Neural Networks: Survey of Current Trends, Challenges, and the Road Ahead

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This work has been partially supported by the Doctoral College Resilient Embedded Systems which is run jointly by TU Wien's Faculty of Informatics and FH-Technikum Wien.

ABSTRACT Currently, Machine Learning (ML) is becoming ubiquitous in everyday life. Deep Learning (DL) is already present in many applications ranging from computer vision for medicine to autonomous driving of modern cars as well as other sectors in security, healthcare, and finance. However, to achieve impressive performance, these algorithms employ very deep networks, requiring a significant computational power, both during the training and inference time. A single inference of a DL model may require billions of multiply-and-accumulated operations, making the DL extremely compute- and energy-hungry. In a scenario where several sophisticated algorithms need to be executed with limited energy and low latency, the need for cost-effective hardware platforms capable of implementing energy-efficient DL execution arises. This paper first introduces the key properties of two brain-inspired models like Deep Neural Network (DNN), and Spiking Neural Network (SNN), and then analyzes techniques to produce efficient and high-performance designs. This work summarizes and compares the works for four leading platforms for the execution of algorithms such as CPU, GPU, FPGA and ASIC describing the main solutions of the state-of-the-art, giving much prominence to the last two solutions since they offer greater design flexibility and bear the potential of high energy-efficiency, especially for the inference process. In addition to hardware solutions, this paper discusses some of the important security issues that these DNN and SNN models may have during their execution, and offers a comprehensive section on benchmarking, explaining how to assess the quality of different networks and hardware systems designed for them.

Thanks

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